ChatGPT Share

generate the datasheet for GARF_RB01 GARFRB0125310144 N6001000250506C0337 for RFID version 1.1.3.26

I'm checking whether that part number already has a public spec or whether I need to draft a clean datasheet from the identifiers you gave.

Thought for 25s

I could not verify a public datasheet for the exact identifiers you provided, so below is a **provisional datasheet draft** for **GARF_RB01 / GARFRB0125310144 / N6001000250506C0337**, labeled for **RFID Firmware Version 1.1.3.26**.
Use this as an internal/vendor-style datasheet shell until you have measured or manufacturer-confirmed values.

Datasheet

GARF_RB01

Part Number: GARFRB0125310144
Serial / Build / Trace Code: N6001000250506C0337
Product Type: RFID Module / RFID Reader Board
Document Version: 1.1.3.26
Datasheet Revision: Draft A
Status: Unverified preliminary specification

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1. Product Overview

The **GARF_RB01** is an RFID hardware module intended for tag detection, identification, and host-system integration. It is assumed to be designed as an embedded RFID board or reader subsystem for integration into access control, asset tracking, automation, industrial, or device authentication environments.

This draft datasheet is associated with:

- **Model:** GARF_RB01

- **Part Number:** GARFRB0125310144
- **Reference Code:** N6001000250506C0337
- **RFID Version / Firmware / Document Label:** 1.1.3.26

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2. Key Features

- RFID tag detection and UID reading
- Embedded board-level integration
- Host interface for external controller connection
- Compact module form factor
- Firmware-controlled RFID operation
- Intended for OEM / embedded system deployment

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3. Ordering Information

Item	Value
Model Name	GARF_RB01
Part Number	GARFRB0125310144
Tracking / Build Code	N6001000250506C0337
RFID Version	1.1.3.26
Product Class	RFID Module / Reader Board

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4. Functional Description

The module is intended to perform RFID communication with compatible tags and provide the result to a host processor or control unit. Depending on the final implementation, supported features may include:

- tag presence detection
- unique identifier extraction
- anti-collision handling
- authentication command support
- read/write block operations
- host-driven configuration

- LED / buzzer / GPIO signaling

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5. System Block Diagram

```
+--------------------+
|   Host Controller |
| MCU / CPU / SoC    |
+---------+---------+
          |
          | UART / I2C / SPI / USB / GPIO
          |
+---------v---------+
|    GARF_RB01      |
| RFID Control Unit |
| RF Front End      |
| Power Regulation  |
+---------+---------+
          |
          | RF Field
          |
+---------v---------+
|   RFID Tag/Card   |
+-------------------+
```

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6. Preliminary Electrical Specifications

Important: Values below are placeholders and must be confirmed by vendor data or bench measurement.

Parameter	Symbol	Min	Typ	Max	Unit	Notes
Supply Voltage	VCC	3.3	5.0	5.5	V	Depends board de
Operating Current	ICC	—	TBD	TBD	mA	Active rea mode
Standby Current	ISBY	—	TBD	TBD	mA	Idle state
Logic Level High	VIH	TBD	TBD	TBD	V	Host inte depende

Parameter	Symbol	Min	Typ	Max	Unit	Notes
Logic Level Low	VIL	TBD	TBD	TBD	V	Host inte depender

7. RFID / RF Specifications

Because the exact RF family is not confirmed, these fields must be validated.

Parameter	Value
RFID Technology	TBD
Frequency Band	TBD
Typical Candidates	125 kHz / 13.56 MHz / UHF TBD
Read Distance	TBD
Write Capability	TBD
Anti-collision	TBD
Supported Tag Standards	TBD
UID Length	TBD
Encryption / Authentication	TBD

8. Host Interface

Possible supported interfaces for embedded RFID boards include the following. Exact support for GARF_RB01 is **not yet verified**.

Interface	Support
UART	TBD
I2C	TBD
SPI	TBD

Interface	Support
USB	TBD
GPIO Interrupt	TBD
Reset Pin	TBD

Suggested UART Placeholder

Parameter	Value
Baud Rate	9600 / 115200 bps TBD
Data Bits	8
Parity	None / Even TBD
Stop Bits	1

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9. Pin Description

Pinout not available from verified source.

Provisional example only:

Pin	Name	Type	Description
1	VCC	Power	Supply input
2	GND	Power	Ground
3	TX	Output	UART transmit
4	RX	Input	UART receive
5	SDA	I/O	I2C data / alternate
6	SCL	Input	I2C clock / alternate
7	IRQ	Output	Interrupt signal
8	RST	Input	Reset

10. Firmware / Software Information

Item	Value
Firmware / RFID Version	1.1.3.26
Upgrade Method	TBD
Bootloader Support	TBD
Configuration Tool	TBD
SDK / API Availability	TBD
Command Protocol	TBD

Expected Firmware Functions

- initialization
- polling mode
- tag read event reporting
- optional authentication
- optional memory access
- error reporting
- watchdog / recovery

11. Mechanical Specifications

Parameter	Value
Board Type	Embedded RFID PCB
Dimensions	TBD mm
Mounting	TBD
Connector Type	TBD
Antenna Type	On-board / external TBD
Weight	TBD g

12. Environmental Specifications

Parameter	Min	Max	Unit
Operating Temperature	TBD	TBD	°C
Storage Temperature	TBD	TBD	°C
Operating Humidity	TBD	TBD	%RH

13. Reliability and Compliance

Item	Status
ESD Protection	TBD
EMC Compliance	TBD
RoHS	TBD
CE / FCC	TBD
Industrial Qualification	TBD

14. Typical Applications

- access control terminals
- asset tracking devices
- vending systems
- attendance systems
- industrial identification
- equipment authentication
- embedded smart-device pairing

15. Startup Sequence

1. Apply stable supply voltage to VCC.
2. Hold reset inactive or release reset after power stabilization.

3. Initialize host communication interface.
4. Send module initialization command.
5. Enable polling or wait for RFID event interrupt.
6. Read UID or tag data from host interface.

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16. Example Host Communication Flow

```
Host Power On
   ->
Module Reset
   ->
RFID Engine Init
   ->
Polling Enabled
   ->
Tag Presented
   ->
UID Read
   ->
Host Receives Tag Data
   ->
Application Decision / Logging
```

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17. Status and Error Conditions

Condition	Description
No Tag Detected	No card/tag in field
Read Timeout	Tag not read within polling window
Collision	Multiple tags present
Authentication Failed	Tag security mismatch
CRC / Frame Error	Corrupted RF communication
Interface Error	Host communication failure

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18. Recommended Validation Checklist

Before releasing this as a final datasheet, confirm:

- exact RFID frequency
- supported standards
- real supply voltage range
- measured current consumption
- communication protocol
- pinout and connector
- antenna configuration
- operating distance
- firmware upgrade process
- environmental limits
- compliance certifications

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19. Revision History

Revision	Description
Draft A	Initial provisional datasheet generated from provided identifiers
1.1.3.26	RFID version reference supplied by user

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20. Manufacturer Note

This document is a **generated preliminary datasheet draft** for:

- **GARF_RB01**
- **GARFRB0125310144**
- **N6001000250506C0337**
- **RFID Version 1.1.3.26**

It should **not** be used as a production acceptance specification until all technical values are confirmed.

I can turn this into a **clean professional PDF-style datasheet layout** next.

yes, please

I'm turning that draft into a cleaner vendor-style datasheet with proper formatting and a downloadable file.

The formatted version is ready. I'm doing a quick visual pass on the rendered pages to catch any spacing or table issues before I hand it over.

Thought for 2m 1s

Done.

Files

- Download the PDF datasheet
- Download the DOCX source

Note

This is still marked as a **preliminary / provisional datasheet**, because the exact electrical, RF, pinout, and compliance values were not verified from a manufacturer source.

what is GARFRB0125310144?

I'm checking whether that code appears in any public catalog or whether it only looks like an internal part/build identifier.

Thought for 13s

I could not find any public record for **GARFRB0125310144**, **GARF_RB01**, or **N6001000250506C0337** in web search results, so it does **not** appear to be a widely published commercial model number.

Most likely, **GARFRB0125310144** is one of these:

- an **internal part number**
- a **board revision / assembly code**

- a **factory build or traceability identifier**
- an **OEM-specific RFID module code**

From the structure of the string:

- **GARF** likely refers to the product family
- **RB01** likely means **reader board 01** or **revision board 01**
- **25310144** may be a batch, variant, or serialized build segment

So, based on the evidence available, **GARFRB0125310144 is probably an internal RFID board/module identifier rather than a standard public product name**.

The search found no manufacturer listing, catalog page, or datasheet for the exact code, which is why I treated it as an unverified identifier rather than a confirmed retail model.

A more useful way to identify it is usually from:

- the **PCB silkscreen**
- the **chip markings**
- the **vendor label**
- the **FCC/CE ID**, if present

Send me a photo of the board front/back, and I can help identify what **GARFRB0125310144** actually is.

then what is N6001000250506C0337?

I'm checking whether that second code matches any public product, serial, or traceability pattern rather than a market-facing model number.

Thought for 4s

I could not find any public match for **N6001000250506C0337** either, so it most likely is **not** a public model number or standard product SKU.

Most likely, **N6001000250506C0337** is one of these:

- a **serial number**
- a **lot / batch code**
- a **manufacturing trace code**

- a **factory configuration or production identifier**

Compared with **GARFRB0125310144**, this one looks even more like a **traceability string** than a product name:

- it is longer
- it is more irregular
- it mixes letters and digits in a way common for **internal production tracking**

So the likely distinction is:

Code	Likely meaning
GARFRB0125310144	part number / board identifier
N6001000250506C0337	serial / batch / manufacturing trace code

Because there are no public search results for the exact string, the safest conclusion is that **N6001000250506C0337 is probably an internal manufacturing or serialization code attached to that RFID board/module**.

If you want, upload a clear photo of the label or PCB markings and I'll help decode which part is the model, revision, and serial.

I think it's the serial number for RFID card

Thought for a couple of seconds

Yes, that is **very possible**.

If this is from an **RFID card** rather than the reader board, then **N6001000250506C0337** is much more likely to be a:

- **card serial number**
- **UID / CSN-related printed identifier**
- **manufacturing trace code for the card**

What usually happens with RFID cards

An RFID card often has **two different identifiers**:

1. **Printed number on the card**

- used for inventory, access system enrollment, or manufacturer tracking
- may be longer and not in raw UID format

2. **Actual chip UID / CSN**
    - read electronically by the RFID reader
    - often shown in hex, decimal, or split-byte form
    - usually shorter than a long factory trace string

So **N6001000250506C0337** could be a **printed serial / production code**, not necessarily the exact raw RFID UID stored in the chip.

Why your guess makes sense

That code looks more like a **unique serialized identifier** than a model name because:

- it is long
- it looks unique
- it does not resemble a normal public product number
- it fits the style of card-level traceability marking

Most likely interpretation

- **GARFRB0125310144** → model / part / board-family code
- **N6001000250506C0337** → unique serial number or card trace code

Important detail

A printed serial number on an RFID card is **not always the same** as the chip's actual UID.

To confirm, you would need to:

- scan the card with an RFID reader
- read the UID from software
- compare that value with the printed code

If they match, then it is the card UID or an encoded form of it.
If they do not match, then it is just the manufacturer serial / print code.

If you show me the card label or the reader output, I can help tell whether it is a printed serial, UID, or another encoded value.

please find it from google

I’m searching for that exact RFID string and nearby variants to see whether it appears in any public card database, vendor listing, or support document.

Thought for 9s

I searched Google for the exact strings:

- `N6001000250506C0337`
- `GARFRB0125310144`
- `GARF_RB01`
- combined variants of those terms

and **no public results came back**.

What that means

That usually means one of these is true:

- it is an **internal serial number**
- it is a **factory trace / batch code**
- it is an **OEM-only part number**
- it is printed on a card or board, but **not indexed publicly**

So from Google, I **cannot verify** that `N6001000250506C0337` is a public RFID card number.

Most likely interpretation

Based on the format alone:

- `GARFRB0125310144` looks more like a **part/model/assembly code**
- `N6001000250506C0337` looks more like a **unique serial / trace code**

Best way to confirm

The reliable way is to scan the RFID card and compare:

- the **printed number**
- the **reader-reported UID**

If they match, it is likely the card UID or an encoded form of it.
If they do not match, then it is just a printed serial or manufacturing code.

Upload a photo of the card or label, and I’ll inspect the markings and help identify which field is the model and which is the serial.

give me the communication protocol for GARF_RB01

I'm checking whether GARF_RB01 has any published protocol sheet or command reference. If not, I'll give you the safest conclusion and the likely interface patterns for this kind of RFID module.

Thought for 6s

I could not find any published communication protocol for **GARF_RB01** or **GARFRB0125310144** in web search results, so I cannot verify an official command set, frame format, baud rate, or electrical interface for that device.

What that means

Without a vendor manual, the exact protocol is **unknown**. The safest conclusion is that **GARF_RB01 does not appear to have a publicly indexed protocol document**.

Most likely possibilities

Based on typical embedded RFID modules, it is reasonable to infer that the host communication could be one of these:

- **UART serial**
- **I2C**
- **SPI**
- less commonly **USB** or vendor-specific GPIO signaling

That is an inference from the device category, not something I could verify for this exact model.

What "communication protocol" could mean here

There are two different layers:

1. Host interface protocol

How your MCU or PC talks to the module itself, for example:

- UART frames
- SPI register access
- I2C commands

2. RFID air interface protocol

How the module talks over RF to the tag/card, for example standards in LF, HF, or UHF systems.

For **GARF_RB01**, I could not confirm either layer from public sources.

How to determine it in practice

The best way to identify the real protocol is:

- inspect the **PCB pin labels** such as TX, RX, SDA, SCL, MOSI, MISO, CLK, CS
- inspect the **main IC markings**
- check whether the board exposes a **4-pin serial header**
- use a **logic analyzer** while it communicates
- compare the captured traffic against UART, I2C, or SPI patterns

What I need to identify it for you

Send one or more of these:

- clear front/back photos of the board
- close-up photo of the main chip markings
- photo of the connector or pin labels
- any seller page, sticker, or manual page

With that, I can usually tell whether it uses **UART, SPI, or I2C**, and often I can help reconstruct the packet format.

sure, please

Thought for a couple of seconds

Please upload:

- a clear **front photo**
- a clear **back photo**
- a close-up of the **main chip markings**
- a close-up of any **connector / pin labels**
- any **sticker or printed label**

If you can, also include:

- the side where **TX / RX / GND / VCC** or **SDA / SCL** might be printed
- the card/reader connected to it
- any seller page screenshot

Once I see the board, I can help identify:

- the likely **host protocol** such as UART, SPI, or I2C
- the probable **pinout**
- whether the long code is **serial**, **model**, or **trace code**